

Project 2

Mark Wang

Contents

(a)	1
(b)	1
(c)	2
(d)	2
(e)	3
(f) and (g)	3
(h)	4
Appendix: full code	5

Same data as Project 1: monthly adjusted close prices for 30 stocks plus the S&P 500, downloaded from Yahoo Finance through the Shiny app, first five years only (January 2017 – December 2021). Short sales are allowed throughout, so the only constraint on the weights is $\mathbf{1}'x = 1$.

Following Merton (1972), Σ is the 30×30 variance-covariance matrix of the stock returns and $\bar{R}$ the 30×1 vector of mean returns.

(a)

$$A = \mathbf{1}'\Sigma^{-1}\bar{R}, \quad B = \bar{R}'\Sigma^{-1}\bar{R}, \quad C = \mathbf{1}'\Sigma^{-1}\mathbf{1}, \quad D = BC - A^2$$

Quantity	Value
A	36.9864
B	1.2563
C	2330.0285
D	1559.1946

$B > 0$ and $C > 0$ because they are quadratic forms of the positive definite Σ^{-1} , and $D > 0$ follows (Merton, footnotes 4 and 5). All three hold here, so the frontier is well defined.

These constants already pin down the global minimum variance portfolio, which sits at $\bar{R}_{pmin} = A/C$ with variance $\sigma_{pmin}^2 = 1/C$:

```
#> A/C          = 0.015874  (1.59% per month)
```

```
#> sqrt(1/C) = 0.020717  (2.07% per month)
```

This matches the minimum risk portfolio computed directly in Project 1 part (f), which is a useful check: the same portfolio arrives by two different routes.

(b)

The Lagrange multipliers come from Merton equation (8):

$$\lambda_1 = \frac{CE - A}{D}, \quad \lambda_2 = \frac{B - AE}{D}$$

Taking $E = 2$ per month, which is above A/C and therefore on the efficient half of the frontier:

```
#> lambda1 = (CE - A)/D = 0.006166
```

```
#> lambda2 = (B - AE)/D = 0.000331
```

(c)

The weights of the frontier portfolio with prescribed return E are

$$x = \Sigma^{-1} [\lambda_1 \bar{R} + \lambda_2 \mathbf{1}]$$

```
#> E      = 0.0200  (2.00% per month)
```

```
#> sd     = 0.0213  (2.13%)
```

```
#> sum of weights = 1.000000, 15 short positions
```

Table 1: Five largest and five smallest weights

Stock	Weight
JPM	0.8742
NFLX	0.2152
VZ	0.1947
LLY	0.1876
AAPL	0.1873
AMD	-0.0882
V	-0.0921
MS	-0.2834
JNJ	-0.3054
BAC	-0.5150

The same vector can be written $x = g + hE$ with

$$g = \frac{1}{D} [B\Sigma^{-1}\mathbf{1} - A\Sigma^{-1}\bar{R}], \quad h = \frac{1}{D} [C\Sigma^{-1}\bar{R} - A\Sigma^{-1}\mathbf{1}]$$

```
#> max |x - (g + hE)| = 5.55e-16
```

```
#> sum(g) = 1.000000  (equals (BC - A^2)/D = 1)
```

```
#> sum(h) = 3.28e-15  (equals (CA - AC)/D = 0)
```

Because $\sum g_i = 1$ and $\sum h_i = 0$, every $x = g + hE$ is automatically fully invested whatever E is chosen. This is what makes part (h) work.

(d)

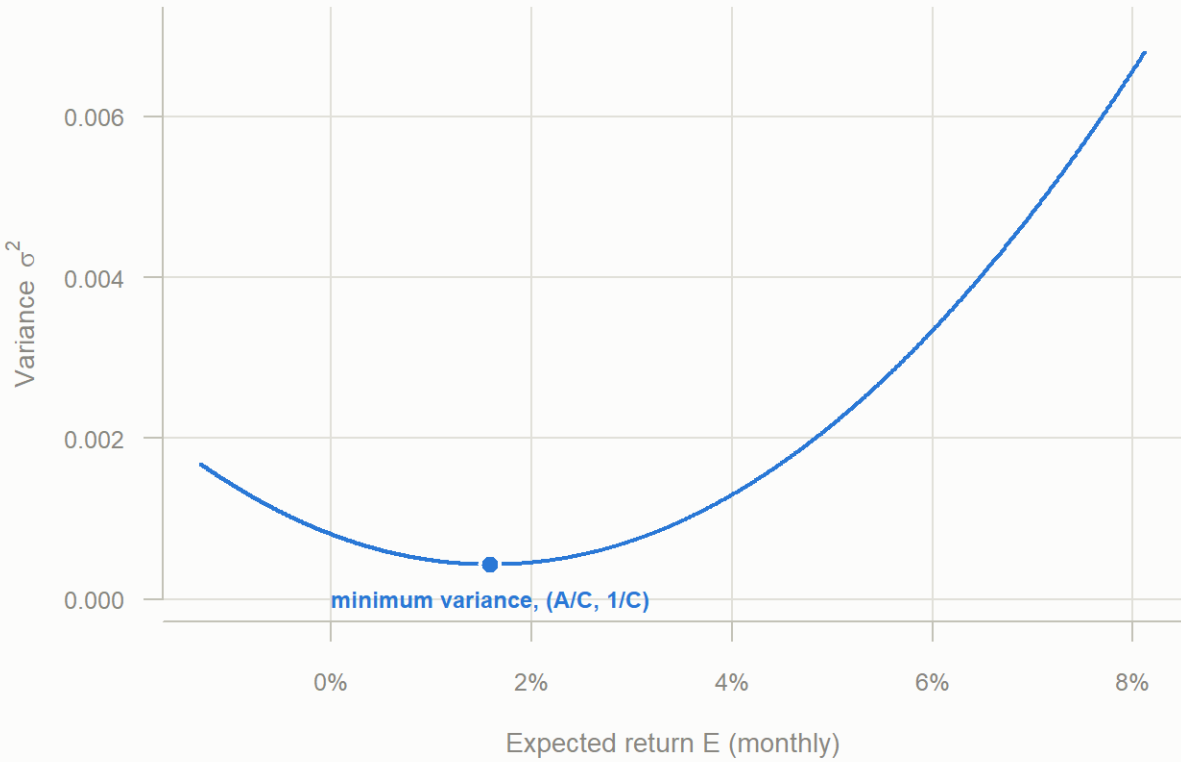
Substituting the multipliers back gives the variance of a frontier portfolio as a function of its expected return,

$$\sigma^2 = \frac{CE^2 - 2AE + B}{D}$$

a parabola in E , with its minimum at $E = A/C$.

(d) Frontier in mean-variance space

$\sigma^2 = (CE^2 - 2AE + B)/D$ - a parabola in E



(e)

Completing the square moves the same relation into mean-standard deviation space:

$$\sigma^2 = \frac{1}{C} + \frac{C}{D} \left(E - \frac{A}{C} \right)^2 \quad \Longleftrightarrow \quad \frac{\sigma^2}{1/C} - \frac{(E - A/C)^2}{D/C^2} = 1$$

which is a hyperbola with centre $(0, A/C)$, vertex at $(\sqrt{1/C}, A/C)$, and asymptotes $E = A/C \pm \sqrt{D/C} \sigma$.

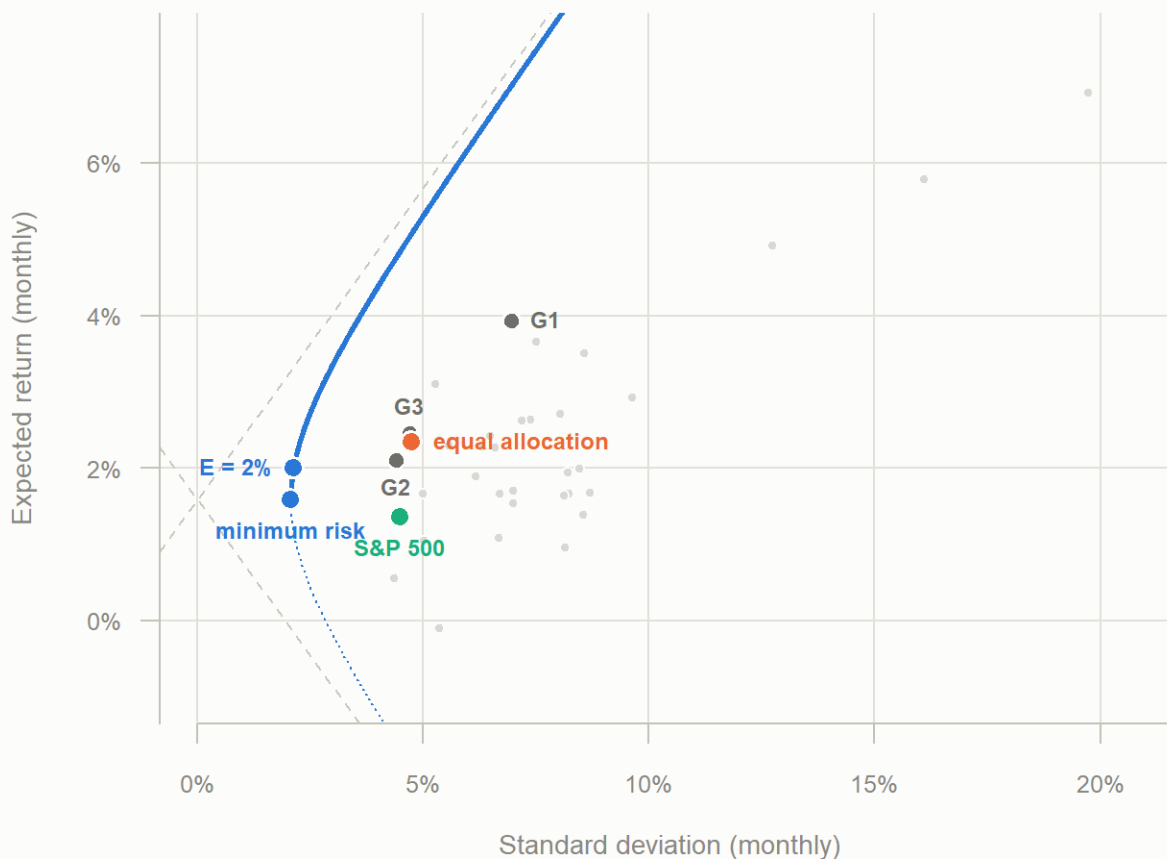
#> max |hyperbola - sqrt(parabola)| over 1200 values of E: 1.39e-17

(f) and (g)

The plot below is the hyperbola from (e) with everything added: the 30 stocks and the S&P 500 in grey and green, the equal allocation and minimum risk portfolios from Project 1, the portfolio from part (c), and three arbitrary portfolios G1–G3 whose weights sum to one (G1 is equal weight on the six technology stocks, G2 random and all long, G3 random with short positions).

(e)-(g) Frontier in mean-standard deviation space

Hyperbola, with the 30 stocks, the index, and four portfolios. Dashed lines are the asymptotes.



Portfolio	Mean (%)	Std. dev. (%)
G1	3.930	6.948
G2	2.104	4.398
G3	2.456	4.707

Every individual stock and every arbitrary portfolio lies to the right of the curve, as it must: the frontier is by construction the smallest standard deviation attainable at each level of expected return. The dotted lower arm is the inefficient half — for each of those portfolios there is one directly above it with the same risk and a higher return.

(h)

By the mutual fund theorem the entire frontier is spanned by any two frontier portfolios. Taking fund 1 to be the portfolio from part (c) and fund 2 a second efficient portfolio with $E = 3.5$, and combining them as in handout #4 page 5,

$$\bar{R}_p = a\bar{R}_1 + (1 - a)\bar{R}_2, \quad \sigma_p^2 = a^2\sigma_1^2 + (1 - a)^2\sigma_2^2 + 2a(1 - a)\sigma_{12}$$

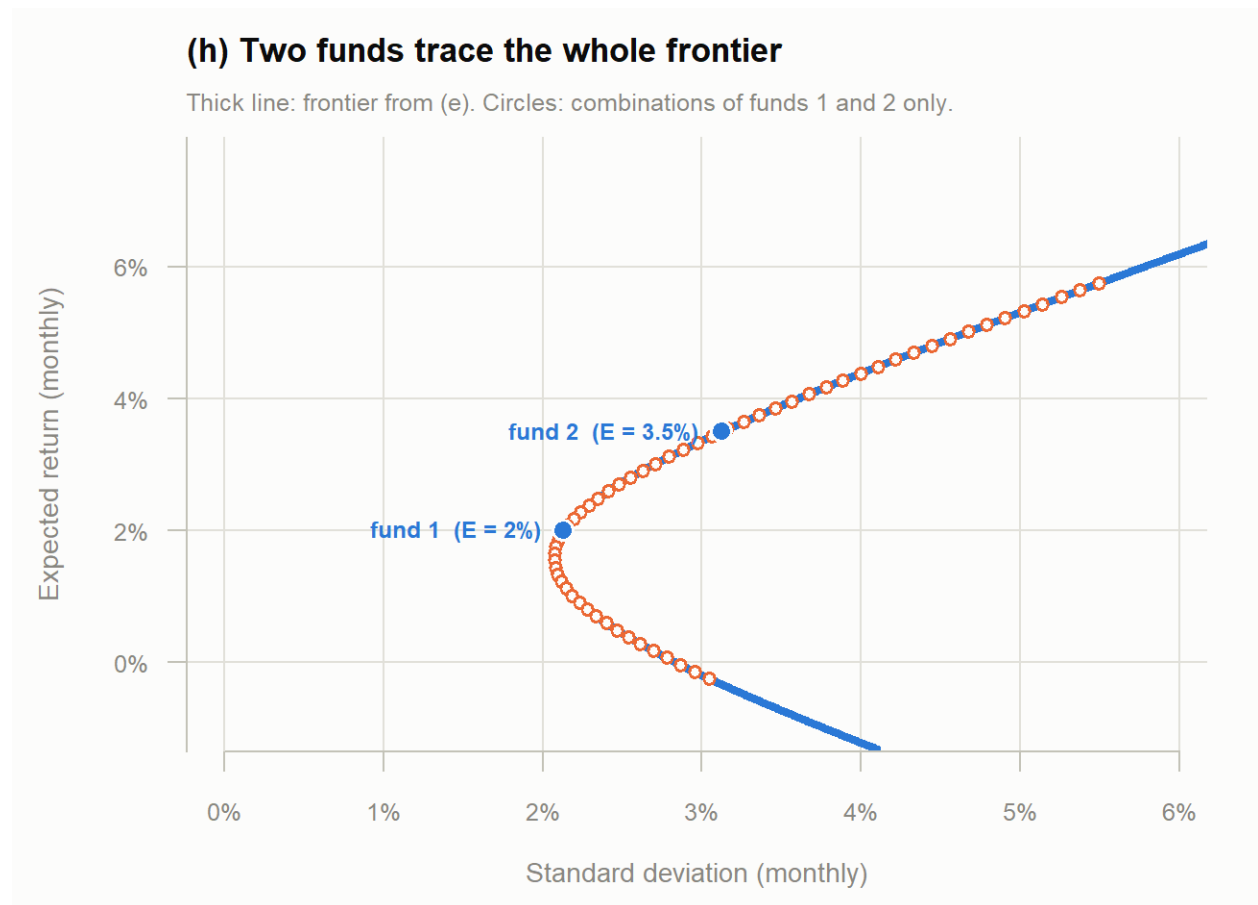
#> fund 1: E = 0.0200, sd = 0.0213

#> fund 2: E = 0.0350, sd = 0.0312

```
#>
#> cov(1,2) from x1' Sigma x2 = 0.00054711
#> cov(1,2) from (C/D)(E1-A/C)(E2-A/C) + 1/C = 0.00054711
```

The second expression is the closed form for the covariance of two frontier portfolios from handout #11 part (c); the two agree exactly, which confirms both funds really are on the frontier.

Letting a range over -1.5 to 2.5 (values outside $[0, 1]$ mean shorting one fund to lever the other) traces the curve below. The circles are combinations of the two funds only — no individual stock is used — and they fall on the frontier from part (e):



```
#> max |two-fund combination - frontier| over 800 mixes: 1.18e-16
```

Agreement to machine precision, so the result is the same as part (e) as expected. The practical reading is that an investor needs only these two funds: any point on the frontier, including the minimum risk portfolio, is some mix of them.

Appendix: full code

```
local({
  f <- sub("^--file=", "", grep("^--file=", commandArgs(FALSE), value = TRUE))
  if (length(f) == 1L && basename(f) == "frontier.R")
    setwd(dirname(normalizePath(f)))
})

DATA <- ".."
```

```

# ---- data -----

a <- read.csv(file.path(DATA, "stockData_train.csv"), sep = ",", header = TRUE,
              check.names = FALSE)
stopifnot(!any(duplicated(names(a))))

prices <- a[, -1]
dates   <- as.Date(prices$Date)
P       <- as.matrix(prices[, -1])
stopifnot(nrow(P) == 60, ncol(P) == 31, !anyNA(P))

n <- nrow(P)
ret <- P[-1, ] / P[-n, ] - 1           # 59 x 31 simple monthly returns

stk     <- setdiff(colnames(ret), "^GSPC")
ret30   <- ret[, stk]
Rbar    <- colMeans(ret30)             # 30 x 1 mean vector
S       <- cov(ret30)                  # 30 x 30 variance-covariance matrix
stdev   <- sqrt(diag(cov(ret)))       # all 31, for the plot
means   <- colMeans(ret)

stopifnot(ncol(S) == 30, qr(S)$rank == 30)

# ---- (a) A, B, C, D -----
#  $A = 1' S^{-1} Rbar$ ,  $B = Rbar' S^{-1} Rbar$ ,  $C = 1' S^{-1} 1$ ,  $D = BC - A^2$ 

Sinv <- solve(S)
one  <- rep(1, 30)

A <- as.numeric(t(one) %*% Sinv %*% Rbar)
B <- as.numeric(t(Rbar) %*% Sinv %*% Rbar)
C <- as.numeric(t(one) %*% Sinv %*% one)
D <- B * C - A^2

stopifnot(B > 0, C > 0, D > 0)           # Merton footnotes 4 and 5

cat("(a) A =", format(A, digits = 6), "\n")
cat("    B =", format(B, digits = 6), "\n")
cat("    C =", format(C, digits = 6), "\n")
cat("    D = BC - A^2 =", format(D, digits = 6), "\n")

# The global minimum variance portfolio sits at  $E = A/C$  with variance  $1/C$ 
E_gmv <- A / C
sd_gmv <- sqrt(1 / C)
cat(sprintf("=> global min variance portfolio: E = A/C = %.6f (%.2f%%), ",
            E_gmv, E_gmv * 100))
cat(sprintf("sd = sqrt(1/C) = %.6f (%.2f%%)\n\n", sd_gmv, sd_gmv * 100))

# ---- (b) Lagrange multipliers -----

EO <- 0.02           # chosen prescribed return: 2% per month, above A/C so efficient
stopifnot(EO > E_gmv)

```

```

lam1 <- (C * E0 - A) / D
lam2 <- (B - A * E0) / D

cat(sprintf("(b) choosing E = %.4f (0.0f%% per month)\n", E0, E0 * 100))
cat("    lambda1 = (CE - A)/D =", format(lam1, digits = 6), "\n")
cat("    lambda2 = (B - AE)/D =", format(lam2, digits = 6), "\n\n")

# ---- (c) composition of the efficient portfolio with return E ----
# x = S^-1 [lambda1 Rbar + lambda2 1]

x_E <- Sinv %*% (lam1 * Rbar + lam2 * one)

# Cross-check against the g + hE form from the Results handout, part (a)
g <- as.numeric(Sinv %*% (B * one - A * Rbar) / D)
h <- as.numeric(Sinv %*% (C * Rbar - A * one) / D)
x_gh <- g + h * E0

stopifnot(all.equal(as.numeric(x_E), x_gh))      # the two forms agree
stopifnot(abs(sum(g) - 1) < 1e-9)               # sum(g) = (BC - A^2)/D = 1
stopifnot(abs(sum(h)) < 1e-9)                  # sum(h) = (CA - AC)/D = 0
stopifnot(abs(sum(x_E) - 1) < 1e-9)            # fully invested
stopifnot(abs(as.numeric(t(x_E) %*% Rbar) - E0) < 1e-12) # hits E

sd_E <- sqrt(as.numeric(t(x_E) %*% S %*% x_E))

cat("(c) efficient portfolio for E =", E0, "\n")
cat(sprintf("    mean = %.6f (0.2f%%)    sd = %.6f (0.2f%%)\n",
            E0, E0 * 100, sd_E, sd_E * 100))
cat(sprintf("    weights sum to %.6f, range %.3f to %.3f, %d short positions\n",
            sum(x_E), min(x_E), max(x_E), sum(x_E < 0)))
cat("    largest 5 weights:\n")
print(round(sort(setNames(as.numeric(x_E), stk), decreasing = TRUE)[1:5], 4))
cat("\n")

# Sanity: the closed form sigma^2 = (CE^2 - 2AE + B)/D must match x'Sx
stopifnot(abs(sd_E^2 - (C * E0^2 - 2 * A * E0 + B) / D) < 1e-14)

# ---- portfolios carried over from Project 1 ----

w_eq <- rep(1 / 30, 30)
mu_eq <- as.numeric(t(w_eq) %*% Rbar)
sd_eq <- sqrt(as.numeric(t(w_eq) %*% S %*% w_eq))

x_min <- Sinv %*% one / C                      # = S^-1 1 / (1' S^-1 1)
mu_min <- as.numeric(t(x_min) %*% Rbar)
sd_min <- sqrt(as.numeric(t(x_min) %*% S %*% x_min))
stopifnot(abs(mu_min - E_gmv) < 1e-12,
          abs(sd_min - sd_gmv) < 1e-12)      # agrees with A/C and sqrt(1/C)

# ---- (g) three arbitrary portfolios ----

set.seed(183)
arb <- list(

```

```

"G1" = local({ w <- numeric(30); w[1:6] <- 1 / 6; w }),      # technology only
"G2" = local({ w <- runif(30); w / sum(w) }),                # random, all long
"G3" = local({ w <- 1 / 30 + rnorm(30, 0, 0.05); w / sum(w) }) # random long/short
)
arb_mu <- vapply(arb, function(w) as.numeric(t(w) %*% Rbar), numeric(1))
arb_sd <- vapply(arb, function(w) sqrt(as.numeric(t(w) %*% S %*% w)), numeric(1))
stopifnot(all(abs(vapply(arb, sum, numeric(1)) - 1) < 1e-12))

cat("(g) three arbitrary portfolios (weights sum to 1):\n")
print(round(data.frame(mean_pct = arb_mu * 100, sd_pct = arb_sd * 100), 3))
cat("\n")

# ---- frontier curves -----

Egrid <- seq(min(means) - 0.012, max(means) + 0.012, length.out = 1200)
var_par <- (C * Egrid^2 - 2 * A * Egrid + B) / D
sd_hyp <- sqrt(1 / C + (C / D) * (Egrid - E_gmv)^2)
stopifnot(max(abs(sd_hyp - sqrt(var_par))) < 1e-12) # same curve, two algebras

# ---- plotting helpers -----

surface <- "#fcbcfb"; ink <- "#0b0b0b"; muted <- "#898781"; grid <- "#e1e0d9"
c_front <- "#2a78d6" # frontier and the portfolios that sit on it
c_eq <- "#eb6834" # equal allocation
c_spx <- "#1baf7a" # S&P 500
c_arb <- "#6f6d67" # arbitrary portfolios

open_png <- function(f, h = 4.7) {
  png(f, width = 6.5, height = h, units = "in", res = 200)
  par(bg = surface, mar = c(4.2, 4.6, 3.4, 1.4), family = "sans")
}

ax <- function(side, at, lab) {
  axis(side, at = at, labels = lab, col = "#c3c2b7", col.axis = muted,
        cex.axis = 0.75, lwd = 1, las = if (side == 2) 1 else 0)
}

# ---- (d) parabola in mean-variance space -----

open_png("fig_parabola.png")
xr <- range(Egrid); yr <- c(0, max(var_par) * 1.04)
plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = pretty(xr), col = grid, lwd = 1)
lines(Egrid, var_par, col = c_front, lwd = 2.2)
points(E_gmv, 1 / C, pch = 21, bg = c_front, col = surface, lwd = 2, cex = 1.5)
text(E_gmv, 1 / C, "minimum variance, (A/C, 1/C)", pos = 1, offset = 0.7,
      cex = 0.72, font = 2, col = c_front)
ax(1, pretty(xr), sprintf("%.0f%%", pretty(xr) * 100))
ax(2, pretty(yr), format(pretty(yr), digits = 2))
mtext("Expected return E (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext(expression(paste("Variance ", sigma^2)), 2, line = 3.2, col = muted, cex = 0.85)
mtext("(d) Frontier in mean-variance space", 3, line = 1.8, adj = 0,
      col = ink, cex = 1.05, font = 2)
mtext(expression(paste(sigma^2, " = (C", E^2, " - 2AE + B)/D - a parabola in E")),

```



```

    3, line = 0.5, adj = 0, col = muted, cex = 0.8)
dev.off()

# ---- (e), (f), (g) hyperbola in mean-sd space -----

open_png("fig_frontier.png", h = 5.2)
is_g <- colnames(ret) == "^GSPC"
xr <- c(0, max(stdev) * 1.06)
yr <- c(min(means) - 0.009, max(means) + 0.007)

plot(NA, xlim = xr, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = pretty(xr), col = grid, lwd = 1)

# asymptotes  $E = A/C \pm \sqrt{D/C} \sigma$  (Merton eq. 16)
abline(a = E_gmv, b = sqrt(D / C), col = "#c9c8c0", lwd = 1, lty = 2)
abline(a = E_gmv, b = -sqrt(D / C), col = "#c9c8c0", lwd = 1, lty = 2)

eff <- Egrid >= E_gmv
lines(sd_hyp[!eff], Egrid[!eff], col = c_front, lwd = 1.1, lty = 3) # inefficient
lines(sd_hyp[ eff], Egrid[ eff], col = c_front, lwd = 2.4)         # efficient

points(stdev[!is_g], means[!is_g], pch = 21, bg = "#d9d8d2",
       col = surface, lwd = 1, cex = 0.8)

pt <- function(x, y, col, lab, pos = 4, cex = 0.72, off = 0.6) {
  points(x, y, pch = 21, bg = col, col = surface, lwd = 1.8, cex = 1.5)
  text(x, y, lab, pos = pos, offset = off, cex = cex, font = 2, col = col)
}

points(arb_sd, arb_mu, pch = 21, bg = c_arb, col = surface, lwd = 1.6, cex = 1.3)
text(arb_sd, arb_mu, names(arb), pos = c(4, 1, 3), offset = 0.5, cex = 0.7,
     font = 2, col = c_arb)
pt(stdev[is_g], means[is_g], c_spx, "S&P 500", 1)
pt(sd_eq, mu_eq, c_eq, "equal allocation", 4)
pt(sd_min, mu_min, c_front, "minimum risk", 1)
pt(sd_E, E0, c_front, "E = 2%", 2)

ax(1, pretty(xr), sprintf("%.0f%%", pretty(xr) * 100))
ax(2, pretty(yr), sprintf("%.0f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(e)-(g) Frontier in mean-standard deviation space", 3, line = 1.8,
      adj = 0, col = ink, cex = 1.05, font = 2)
mtext("Hyperbola, with the 30 stocks, the index, and four portfolios. Dashed lines are the asymptotes.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.72)
dev.off()

# ---- (h) mutual fund theorem -----
# Two efficient portfolios span the whole frontier. Take the one from (c) and a
# second efficient portfolio, then combine them as in handout #4 page 5:
#  $E_p = a E_1 + (1-a) E_2$ 
#  $sd_p^2 = a^2 s_1^2 + (1-a)^2 s_2^2 + 2 a (1-a) s_{12}$ 

E1 <- E0; x1 <- as.numeric(x_E)

```

```

E2 <- 0.035; lam1b <- (C * E2 - A) / D; lam2b <- (B - A * E2) / D
x2 <- as.numeric(Sinv %*% (lam1b * Rbar + lam2b * one))
stopifnot(abs(sum(x2) - 1) < 1e-9, abs(as.numeric(t(x2) %*% Rbar) - E2) < 1e-12)

s1 <- sqrt(as.numeric(t(x1) %*% S %*% x1))
s2 <- sqrt(as.numeric(t(x2) %*% S %*% x2))
s12 <- as.numeric(t(x1) %*% S %*% x2)

# The Results handout, part (c), gives the covariance of two frontier
# portfolios in closed form. It must agree with x1' S x2.
s12_closed <- (C / D) * (E1 - A / C) * (E2 - A / C) + 1 / C
stopifnot(abs(s12 - s12_closed) < 1e-14)

agrid <- seq(-1.5, 2.5, length.out = 800)
mf_E <- agrid * E1 + (1 - agrid) * E2
mf_sd <- sqrt(agrid^2 * s1^2 + (1 - agrid)^2 * s2^2 + 2 * agrid * (1 - agrid) * s12)

# Every combination must land exactly on the frontier: check sigma against the
# closed form for its own expected return.
stopifnot(max(abs(mf_sd^2 - (C * mf_E^2 - 2 * A * mf_E + B) / D)) < 1e-15)

cat("(h) mutual fund theorem\n")
cat(sprintf("    fund 1: E = %.4f, sd = %.4f    fund 2: E = %.4f, sd = %.4f\n",
            E1, s1, E2, s2))
cat(sprintf("    cov(1,2) = %.8f from x1'Sx2, %.8f from (C/D)(E1-A/C)(E2-A/C)+1/C\n",
            s12, s12_closed))
cat(sprintf("    max |combination - frontier| over %d mixes: %.2e\n",
            length(agrid), max(abs(mf_sd - sqrt((C * mf_E^2 - 2 * A * mf_E + B) / D))))))

open_png("fig_mutualfund.png")
sel <- mf_E >= yr[1] & mf_E <= yr[2]
xr2 <- c(0, max(mf_sd[sel]) * 1.08)
plot(NA, xlim = xr2, ylim = yr, axes = FALSE, xlab = "", ylab = "")
abline(h = pretty(yr), v = pretty(xr2), col = grid, lwd = 1)
lines(sd_hyp, Egrid, col = c_front, lwd = 4)
points(mf_sd[sel][seq(1, sum(sel), by = 14)], mf_E[sel][seq(1, sum(sel), by = 14)],
       pch = 21, bg = surface, col = c_eq, lwd = 1.6, cex = 0.85)
pt(s1, E1, c_front, "fund 1 (E = 2%)", 2)
pt(s2, E2, c_front, "fund 2 (E = 3.5%)", 2)
ax(1, pretty(xr2), sprintf("%.0f%%", pretty(xr2) * 100))
ax(2, pretty(yr), sprintf("%.0f%%", pretty(yr) * 100))
mtext("Standard deviation (monthly)", 1, line = 2.6, col = muted, cex = 0.85)
mtext("Expected return (monthly)", 2, line = 3.2, col = muted, cex = 0.85)
mtext("(h) Two funds trace the whole frontier", 3, line = 1.8, adj = 0,
      col = ink, cex = 1.05, font = 2)
mtext("Thick line: frontier from (e). Circles: combinations of funds 1 and 2 only.",
      3, line = 0.5, adj = 0, col = muted, cex = 0.75)
dev.off()

cat("\nwrote fig_parabola.png, fig_frontier.png, fig_mutualfund.png\n")

```